ST121 - 2026 - Statistical Laboratory
Lab report 1

Important information

Your reports should be submitted electronically on Moodle. The deadline is 1pm on Thursday 19th February. If you have any doubt or problems with submission, let me know as soon as possible, so we can sort that out.

Please note that your report must be submitted in PDF. You should try and limit yourself to about 6/8 sides of A4, but without counting for any figures which might be needed. In particular, do **not** print very long vectors, if they are not required.

Also, please make sure you include your student ID code and lab group number and/or tutor name on the front sheet, but **NOT** your name!

In the initial marking scheme (I might amend it slightly), there will be a total of **60 marks** for this assignment, including **4** marks allocated to the writing layout and general presentation of your report. So make sure to be clear when answering, give the requested answers, show all necessary code, show relevant outputs (especially for sections 3-4), explain by using sentences to link your reasoning, include appropriate justifications, label graphs, make the report nice to read for the marker... I will ask the markers to be fair but not too lenient either on this aspect.

Questions, comments and corrections to Samuel.Touchard@warwick.ac.uk

Exercises

1. Vectors

Give one command/line of code which will generate the following:

(i) `(-8, -5, -2, 1, 4, 7, 10, 13)`

(ii) `(2, 2, 2, 4, 4, 4, 6, 6, 6, 8, 8, 8, 10, 10, 10, 12, 12, 12)`

(iii) `(9, 9, 9, 9, 9, 9, 6, 6, 6, 6, 6, 3, 3, 3, 3, 0, 0, 0, -3, -3, -6)`

(iv) `(1, 8, 81, 1024, ..., 3486784401, 100000000000)`

(v) `(-3, 0, 3, -9, 0, 9, -27, 0, 27, -81, 0, 81, -243, 0, 243)`

(vi) `1 5 9 13 2 6 10 14 3 7 11 15 4 8 12 16`

Note: You are expected to give a command other than the exact representation of a vector. For example, you cannot give `c(-8, -5, -2, 1, 4, 7, 10, 13)` as answer to part (i).

[6 marks]

2. Random numbers and vectors

1. Create a vector `x` which will contain 1000 realisations from an Exponential distribution, with mean 2.

2. Then give the commands to produce:

   (i) a vector containing the 200^{th}, 400^{th}, 600^{th}, 800^{th} and 1000^{th} elements;

   (ii) a vector containing all elements except those from the 1^{st} to the 100^{th} element and those from the 900^{th} to the 1000^{th}; *[Only give the command, DO NOT print the resulting vector!]*

   (iii) the number (or proportion) of elements between 6 and 9;

   (iv) the theoretical answer to (iii);

   (v) a random sample of size 500 with replacement from the elements of `x` which are less than 5.

[6 marks]

3. Short mathematical questions

These questions concern the use of `R` to address mathematical questions. For each problem below include your code in the report as well as answering the questions posed.

1. Prime numbers

   (a) Write an `R` function `primefind` which returns TRUE if its integer argument is prime. Recall that a prime is any number divisible only by itself and 1, and 1 is not a prime.

   (b) Using your previous function, create a new function which accepts a list of integers and returns only the values which are prime.

   (c) List all prime numbers between 1 and 300.

2. Square numbers

   (i) Write a `R` function that checks whether a positive integer is a square number, *i.e.* if its square root is also an integer.

   (ii) Using your previous function, write a new `R` function that has as its input a vector where each element is a positive integer, and which returns each element of the input vector which is a square number. If you haven't managed to get `square.test` working, don't worry; you won't lose marks for this part if you use the function here.

   (iii) How many square numbers do we have between 1000 and 10000 included?

   (iv) The following formula is a result (result you might already know actually) about the sum of consecutive squared numbers:

   $$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6},$$

   so that the sum of the squares of the first n integers is equal to $\frac{n(n+1)(2n+1)}{6}$. Using `R` and the output from the previous question, show that the partial sum of square numbers between 1000 and 10000 verify that result.

[16 marks]

4. Longer mathematical question

4.1. Preliminary questions

(a) Let consider two points P_1 and P_2, of co-ordinates (x_1, y_1) and (x_2, y_2) respectively.
Write a `line.equation` function that takes as inputs the points co-ordinates (x_1, y_1, x_2, y_2) and returns the intercept and slope of the line P_1P_2, as well as the length of the segment.

(b) Let consider two lines, of equation $y = a_1x + b_1$ and $y = a_2x + b_2$. Assuming the two lines are not parallel, write an `intersection` function that returns the co-ordinates of the intersection point of these two lines.

(c) Let consider a triangle, whose sides have lengths a, b, c.
Write a `triangle.area` function that takes as input a vector containing the three lengths, and returns the area of the triangle.
Hint: you may want to use Heron's formula.

[8 marks]

4.2. Triangles in the unit square

(a) For reproducity reasons, set a seed equal to the **last** three digits of your student ID number.

(b) We will consider in total 9 points in/on the unit square.
The first 4 will be the vertices: $O(0, 0), P(1, 0), Q(1, 1), R(0, 1)$.
The next 4 will be on the edges: $A(x_A, 0), B(1, y_B), C(x_C, 1), D(0, y_D)$, where x_A, y_B, x_C and y_D are randomly generated from a uniform distribution.
The 9th point. I, will be defined later.

Create a data frame that contains the points information (name and co-ordinates). You may want to include I already, with `NA` co-ordinates for example, or you may want to include it later.

(c) Create, plot and label the points. Also include, in dashed lines, the edges of the square, i.e. lines of equation $x = 0/1$ and $y = 0/1$.
Hint: you may want to consider the `text` function for labeling the points, and the functions `xline` and `yline` from the `fields` package for the edges. But I am sure other valid alternatives exist.

(d) Plot the lines connecting the vertices of $ABCD$, including the diagonals.
Hint: have a look at the `abline` function.

(e) The lines AC and BD will cross at I. Label it on the graph. Find its co-ordinates and update your data frame of points information.

(f) Write a function `triangle.area.by.name` that takes as input a triangle name (e.g. `"OAD"` or `"APB"`) as a string object and computes its area.
Potential hint: have a look at the help file for the `substring` function. Again, it might not be the only valid solution.

(g) Compute the area of all 8 "elementary" triangles (non-overlapping any other) present in the unit square. Identify the smallest, before returning its name and area.

[20 marks]

Total for the whole assignment: 60 marks